

# Ivan Kashuba

## Research Engineer & R&D Chemist

Materials Science | Polymers & Coatings | Photonics | Process Engineering | Scientific Data Workflows

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## CURRICULUM VITAE

Alanya, Türkiye

### PROFESSIONAL PROFILE

Research engineer and chemist-technologist with 10+ years of laboratory, R&D and production experience in silver-halide single crystals and infrared light guides, optical coatings, glycolide monomer manufacturing and powder coatings. Combines hands-on experimentation and equipment commissioning with instrument-data interpretation, process troubleshooting, engineering calculations and technical documentation. Current independent work focuses on reproducible Python models, laboratory document automation and computational chemistry and bioinformatics foundations.

### RESEARCH AND TECHNICAL FOCUS

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| <b>Materials and process R&amp;D</b><br>Single crystals, infrared light guides, polymers, powder and optical coatings, glycolide, thermal-vacuum processing and extrusion.              | <b>Experimental analysis and troubleshooting</b><br>Instrument data, process parameters and material condition used for root-cause analysis, corrective action and reproducible procedures. |
| <b>Scientific data and engineering calculations</b><br>Excel and Python models, CSV/JSON workflows, plotting, validation checks, literature-based calculations and technical reporting. | <b>Laboratory automation and digital R&amp;D</b><br>Git/GitHub, FastAPI and UiPath workflows, deterministic specification checks, human-review routing and AI-assisted technical work.      |

### PROFESSIONAL EXPERIENCE

#### Independent Research, Engineering Modelling & Professional Development | Independent R&D

Feb 2025 - Present

Alanya, Türkiye

- Conduct independent literature research and maintain a public GitHub portfolio in materials, polymers, coatings, monomer processes and scientific data workflows.
- Convert selected chemical-engineering calculations from Excel into reproducible Python models with configurable inputs, validation checks, CSV outputs and plots.
- Develop and document an AI Laboratory Document Automation project combining FastAPI, SQLite, UiPath C# workflows and deterministic PASS/REVIEW/FAIL logic while strengthening version control and technical writing.

#### Research Engineer - Powder Coatings Laboratory Setup & Process Improvement | Institute for Applied Chemistry and Certification FRISHBERG Ltd.

Jul 2024 - Jan 2025

Yekaterinburg, Russia

- Helped establish and develop a powder-coatings R&D laboratory by transferring, positioning, connecting and commissioning selected production equipment for laboratory use.
- Investigated replacement components and alternative formulations through gel-time measurements, DSC with nitrogen cooling, NMR-solvent selection and laboratory coating trials.
- Processed and visualised Excel data for component content, gel time, powder humidity, cathodic-disbondment diameter, material quantities and correlation screening.
- Improved temperature control and powder quality by inspecting and cleaning a liquid heat exchanger, addressing micronisation cleanliness, improving storage sealing and increasing mixture homogeneity through grinding and sieving.
- Applied tribostatic powder spraying to heated test specimens; produced approximately 30 kg of experimental coating batches and assessed adhesion and cathodic disbondment.

**Chemist-Technologist - Glycolide Monomer Manufacturing & Process****Improvement | Medin-N****Aug 2021 - Apr 2024***Yekaterinburg, Russia*

- Maintained technical process documentation and process sheets for glycolide (1,4-dioxane-2,5-dione) synthesis, purification, filtration and thermal-vacuum operations used in biodegradable polymer production.
- Investigated quality deviations related to raw-material dehydration, storage-associated oxidation, filtration and thermal-vacuum processing; identified ineffective dehydration and supported installation of an improved drying cabinet.
- Defined practical temperature ranges for thermal-vacuum synthesis and distillation and improved filtration by insulating the filter element, reducing overheating and unintended oligomer dissolution.
- Contributed to production of approximately 600 kg of monomer for biodegradable polymer applications.
- Reduced preparation time for a heat-crosslinking polymer-dye protective solution from approximately two weeks to a few hours through grinding, heating and controlled mixing.
- Performed  $^1\text{H}$  NMR analysis and identified rapid moisture-associated monomer oligomerisation; co-developed a microwave synthesis and vacuum-distillation setup for university project work.

**Process Engineer - Optical Coatings | ODV Prescription Optics Plant****Apr 2021 - Jul 2021***Yekaterinburg, Russia*

- Supported washing, surface activation and adhesion processes for polymer ophthalmic lenses and optical interference coatings.
- Transferred an interference-coating process to new low-pressure electron-beam deposition equipment.
- Measured polymer moisture content and coating transmission spectra; maintained Excel process logs and investigated coating defects using microscopy, humidity, temperature and equipment operating parameters.
- Removed hardened flaking polymer contamination from the finishing area and supported rapid corrective action during production.

**Research Engineer - Fiber Technology & Photonics | Fiber Technology and****Photonics Laboratory****Jan 2013 - Jul 2021***Yekaterinburg, Russia*

- Worked in the laboratory where specialist-degree and postgraduate research in infrared light guides, single-crystal materials and photonics was completed.
- Played a key hands-on role in a team that developed and implemented a flexible PMMA light-guide neonatal phototherapy lamp for treatment of hyperbilirubinaemia in premature infants.
- Proposed automated laser engraving of PMMA light guides to create controlled side emission and identified suitable equipment suppliers.
- Co-assembled an LED UV-curing unit for rapid connector fixation; machined connectors and attachment components and supported infrared light-guide processing and extrusion work.
- Prepared a detailed written workflow for a substantial part of lamp production and assembly and participated in volunteer projects supporting children in orphanages.

**SELECTED CONTRIBUTIONS ACROSS ROLES**

| Process improvement                                                                                               | Equipment and prototyping                                                                                                     | Data and documentation                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Drying, filtration, temperature control, micronisation cleanliness, storage sealing and workflow standardisation. | Laboratory commissioning, extrusion tooling, tubular furnace construction, UV-curing unit assembly and prototype manufacture. | Excel analysis, instrument-data interpretation, process sheets, production procedures, GitHub documentation and reproducible calculations. |

## EDUCATION AND ACADEMIC TRAINING

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### Completed Postgraduate Programme in Chemical Sciences | Ural Federal University

2015 - 2019

*Institute of Chemical Engineering | GPA 4.4/5.0*

- Diploma qualification awarded: "Researcher; Teacher-Researcher."
- Research topic: "Research and Development of Photonic-Crystal Light Guides for a Wide Infrared Spectral Range."
- Refined the AgBr-AgI phase diagram using differential thermal analysis and investigated compositions intended for single-mode infrared light guides with a wider transmission range.
- Worked with thermozone crystallisation and Bridgman-Stockbarger crystal growth; machined single crystals in a glove box and hot-extruded them into infrared light guides.
- Designed extrusion tooling in KOMPAS-3D, reviewed ANSYS structural-stress results prepared with a simulation specialist, and calculated and built a tubular furnace for heating extrusion tooling.

### Specialist Degree in Chemical Technology | Ural Federal University

2010 - 2015

*Institute of Chemical Engineering | GPA 4.4/5.0*

- Specialisation: Chemical Technology of Single Crystals, Materials and Electronic Products.
- Graduation research: "Study of Extrusion Modes of Infrared Light Guides Based on Silver-Halide Single Crystals."
- Investigated production and hot-extrusion conditions for two-component silver-halide infrared light guides, including crystal machining, tooling strength and optical-quality evaluation.
- Designed and tested extrusion tooling and calculated a tubular furnace integrated with a Shimadzu AGS-X testing machine for hot extrusion.

### Professional Retraining Diploma - Enterprise Economics and Management | Ural Federal University Business School

2012 - 2015

*Yekaterinburg, Russia*

- Coursework included profitability and cost calculations for a small-scale scientific production process, business planning, operations management and industrial decision-making.

### Additional University Studies - Resource-Saving Technologies and Biotechnology | Ural Federal University

2023 - 2024

*Programme not completed because of international relocation*

- Completed practical coursework in NMR interpretation, chromatographic and spectroscopic methods, molecular docking, protein-ligand interaction analysis, phylogenetic analysis and Excel-based process scale-up equations.
- Studied introductory Python data workflows and literature-based design of a microwave reactor for monomer synthesis and vacuum distillation.
- Tools studied included PyMOL, AutoDockTools, Open Babel, ProteinsPlus, SwissTargetPrediction, RCSB PDB, NCBI and PubMed.

## RESEARCH THEMES DEVELOPED DURING EDUCATION

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**Infrared optical materials:** Silver-halide crystals, solid solutions, phase diagrams, wide-band infrared transmission, single-mode and microstructured light guides.

**Crystal growth and processing:** Thermozone crystallisation, Bridgman-Stockbarger growth, glove-box machining, hot extrusion and tooling/furnace design.

**Chemical process engineering:** Thermal-vacuum synthesis, distillation, filtration, drying, similarity criteria, scale-up calculations and process troubleshooting.

**Biomedical and computational applications:** Laser-surgery light guides, neonatal phototherapy, biodegradable polymers, molecular docking and literature-driven hypothesis development.

## SELECTED RESEARCH AND ENGINEERING PROJECTS

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**Digital and reproducible engineering portfolio** - Public repositories are documented for reproducibility. The calculation projects are educational/theoretical models and are not substitutes for validated industrial design calculations.

### **AI Laboratory Document Automation** [UiPath](#) | [FastAPI](#) | [SQLite](#) | [C#](#) | [Python](#) [Project link](#)

- Built a portfolio MVP that extracts structured values from laboratory documents, validates them against configurable specifications and returns deterministic PASS, REVIEW or FAIL decisions.
- Implemented a FastAPI service, SQLite audit history, CSV export, automated tests and UiPath C# workflows for passing, incomplete and out-of-specification scenarios.
- Designed human-in-the-loop routing so low-confidence, incomplete or out-of-specification results require review; an LLM may summarise results but cannot change the rule-engine decision.

### **Reactor Dissolution Calculation** [Mass transfer](#) | [Mixing](#) | [Energy](#) [Project link](#)

- Parameter-driven Python model for agitator-speed scenarios, mass transfer, dissolution time, motor-capacity limits and energy indicators.
- Baseline outputs were checked against the source Excel calculation and documented with configurable inputs and plots.

### **Pneumatic Dryer Performance Calculator** [Heat and mass transfer](#) | [Pressure drop](#) [Project link](#)

- Evaluates gas velocity, tube diameter, heat transfer, drying time, required pipe length and pressure drop over a configurable operating range.
- Produces structured CSV outputs and visualisations for scenario comparison.

### **Stirred-Vessel Heating Comparator** [Agitation](#) | [Heat transfer](#) [Project link](#)

- Compares paddle, anchor and propeller impellers across rotational-speed ranges, predicted heat-transfer performance, heating time and diminishing-return behaviour.
- Baseline results were checked against the source spreadsheet and documented as a reproducible workflow.

### **Tube-in-Tube Heat Exchanger Scaling Model** [Scale-up](#) | [Similarity criteria](#) [Project link](#)

- Converts an Excel-based milk-heating calculation into a parameter-driven Python workflow with criteria-based flow screening and improved heat-transfer assumptions.
- Uses validation and explicit limitations to separate coursework reproduction from engineering design claims.

## EXPERIMENTAL AND PRODUCT DEVELOPMENT PROJECTS

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### **Flexible Neonatal Phototherapy Lamp** [Medical device development](#) | [PMMA light guides](#)

- Contributed practical manufacturing and process-development work to a flexible side-emitting PMMA light-guide lamp for neonatal phototherapy.
- Proposed laser engraving for controlled side emission, identified equipment suppliers, co-assembled a UV-curing unit, machined connectors and documented assembly procedures.

### **Microwave Synthesis and Vacuum-Distillation Setup** [Monomer synthesis](#) | [Literature-based design](#)

- Co-developed a laboratory setup for microwave-assisted synthesis and vacuum distillation during university project work conducted in collaboration with the monomer laboratory.
- Selected heating media and process concepts through literature review;  $^1\text{H}$  NMR was used to investigate moisture-associated glycolide oligomerisation.

## SELECTED PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

1. Korsakova, E. A.; L'vov, A. E.; **Kashuba, I. A.**; Korsakov, V. S.; Salimgareev, D. D.; Korsakov, A. S.; Zhukova, L. V. "Fiber-Optic Assemblies Based on Polycrystalline Lightguides for the Mid-IR." Journal of Optical Technology 86(7), 439-445 (2019). [doi: 10.1364/JOT.86.000439](https://doi.org/10.1364/JOT.86.000439)
2. Korsakova, E.; **Kashuba, I.**; Korsakov, A.; Doroganov, S.; Skvortsov, D.; Kalintseva, N.; Zhukova, L. "Laser Systems Supplied with Silver Halide Fibres for Laser-Surgery Angioplasty." Proceedings of the 2019 Ural Symposium on Biomedical Engineering, Radioelectronics and Information Technology (USBREIT), 2019. [doi: 10.1109/USBREIT.2019.8736633](https://doi.org/10.1109/USBREIT.2019.8736633)
3. Salimgareev, D. D.; L'vov, A. E.; Korsakov, V. S.; Gul'ko, D. Y.; **Kashuba, I. A.**; Korsakov, A. S.; Zhukova, L. V. "β-Radiation-Resistant Optical Fibers Based on Modified Silver Halide Crystals." Materials of Nano-, Micro-, Optoelectronics and Fiber Optics: Physical Properties and Applications, 17th International Scientific Conference-School, Saransk, 2018, p. 40.

Research profiles: [ORCID 0009-0002-8976-8886](https://orcid.org/0009-0002-8976-8886) | [Ural Federal University publications profile](#)

## TECHNICAL METHODS AND EQUIPMENT

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Materials and processes | Silver-halide single crystals; thermozone crystallisation; Bridgman-Stockbarger growth; glove-box machining; hot extrusion; powder-coating formulation and application; glycolide synthesis; thermal-vacuum processing and distillation; filtration; drying; vacuum deposition; optical-surface cleaning and adhesion; micronisation.        |
| Analytical methods      | Hands-on use and/or interpretation: <sup>1</sup> H NMR, DSC, DTA, TGA, FTIR, UV-Vis/spectrophotometry, microscopy, moisture, pH, conductivity and thickness measurements. Introductory training: <sup>13</sup> C NMR, HSQC, HMBC, TLC, HPLC, LC-MS, GC-MS and UPLC.                                                                          |
| Equipment               | DSC with nitrogen cooling; thermal-vacuum reactors and vacuum systems; electron-beam deposition equipment; powder-coating and micronisation equipment; extrusion tooling and testing machine; CO <sub>2</sub> laser; UV-curing equipment; microwave reactor; ultrasonic bath; centrifuge; laboratory drying, grinding and heating equipment. |
| Quality and test work   | Gel-time measurement; cathodic-disbondment evaluation; adhesion and coating trials; spectral-transmission measurement; experimental batch preparation; process-deviation investigation; procedure development and technical documentation.                                                                                                   |

## SOFTWARE, RESEARCH TOOLS AND DIGITAL SKILLS

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| Established use                  | Microsoft Excel; Origin; MestReNova; ChemDraw; Chem3D; KOMPAS-3D; Microsoft Office; Google Workspace; scientific literature search; technical documentation and presentation development.                                                  |
| Developing use                   | Python and Google Colab for engineering calculations, data processing and visualisation; Git and GitHub; JSON/CSV workflows; UiPath Studio; FastAPI; SQLite; AI-assisted coding, research and documentation workflows.                     |
| Research databases and modelling | Reaxys; Google Scholar; PubMed and NCBI; RCSB PDB; ProteinsPlus; PyMOL; AutoDockTools; Open Babel; SwissTargetPrediction. ANSYS: review and interpretation of structural-analysis results rather than independent expert-level simulation. |

## LANGUAGES

Russian - Native | English - B2, professional working proficiency | Chinese - A1, elementary

Personal website: [ivan-kashuba-rd.github.io](https://ivan-kashuba-rd.github.io) | Full project portfolio and current contact details